



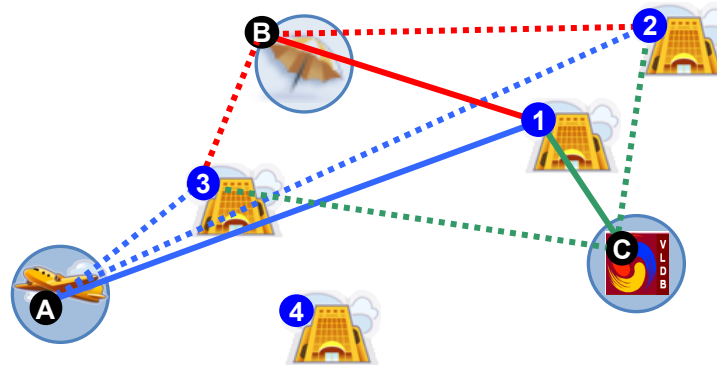
The Spatial Skyline Queries

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Outline

- Motivation
- Problem Definition
- Related Work
- Geometric Properties
- Our Algorithms: VS^2 and B^2S^2
- Performance Evaluation
- Conclusion and Future Work

Motivation

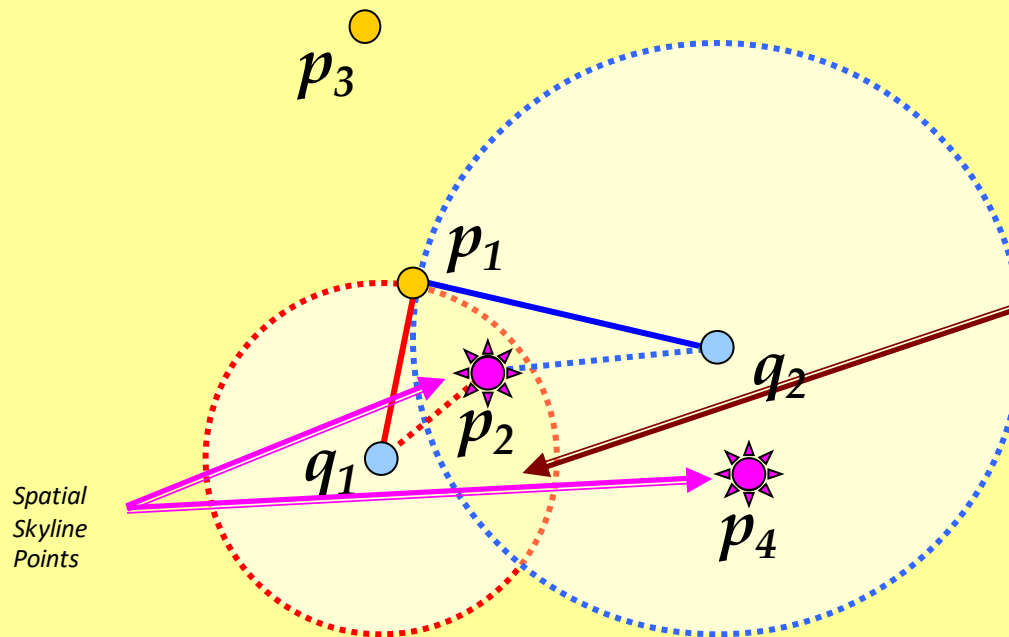


- H1 is better than H2
- H1 is closer than H3 to C but farther than H3 to A
- No hotel is better than H1 or H3 or H4

- **Problem:** Finding Hotels close to Airport, Beach, and Conference
- **Query:** What are the candidate *interesting* hotels?
 - A skyline query with dynamic spatial attributes ...
 - **Criteria for an interesting hotel:** No hotel is *closer* than a candidate hotel to A, B, and C
 - No hotel is better than a candidate hotel in terms of all distances to A, B, and C (i.e., 3 query functions to be optimized together)
- **Applications:** Trip Planning, Crisis Management, Defense and Intelligence, Wireless Sensor Networks

Problem Definition

p_1 **spatially dominates** p_2 with respect to Q iff
 $D(p_1, q_i) \leq D(p_2, q_i)$ for all q_i in Q and
 $D(p_1, q_j) < D(p_2, q_j)$ for at least one q_j



- Data $P = \{p_1, p_2, p_3, p_4\}$
- Query $Q = \{q_1, q_2\}$
- Distance $D() = \text{Euclidean}$
- p_2 **spatially dominates** p_1 with respect to $\{q_1, q_2\}$
- **Dominator Region** of p_1
- p_1 **spatially dominates** p_3
- **Dominance Region** of p_1
- No dominance relation between p_1 and p_4

Spatial Skyline Query (SSQ): find the data points p_i that are **not** spatially dominated by any other point p_j with respect to the given query points (here, p_2 and p_4).

Related Work

• General Skyline Query

- ✓ BNL and D&C, Börzsönyi et al., ICDE'01
- ✓ Bitmap and Index, Tan et al., VLDB'01
- ✓ NN, Kossmann et al., VLDB'02
- ✓ SFS, Chomicki et al., ICDE'03
- ✓ BBS, Papadias et al., SIGMOD'03
- *Static attributes vs. dynamic spatial attributes in SSQ*
- *SSQ is a dynamic skyline query*

Hotel Information
(price, #of rooms)

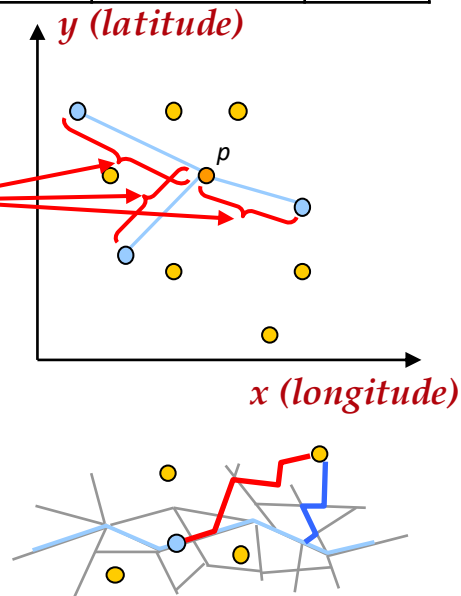
Name	# of rooms	Price
Hotel 1	20	70
Hotel 2	40	40
Hotel 3	40	100
Hotel 4	50	70
Hotel 5	60	100
Hotel 6	70	10
Hotel 7	80	40

• Nearest Neighbor Search

- ANN, Papadias et al., TODS 2005, 30(2)
 - *Looks for subsets of spatial skyline points*

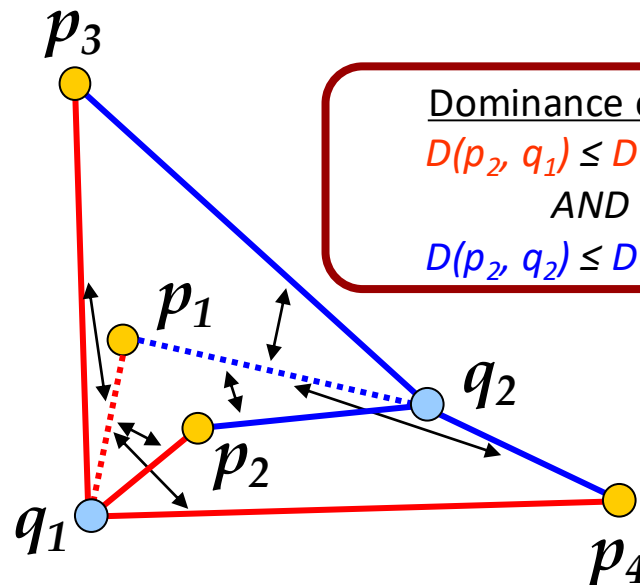
• NN and Skyline

- Huang and Jensen, W2GIS'04
 - *Each point-of-interest has 2 dimensions: minimum distance to query point and minimum detour to pre-defined route → dynamic skyline*
 - *Limited setting*
 - *Uses naïve in-memory skyline computation*



Naïve Solution

- Data $P = \{p_1, p_2, p_3, p_4\}$
- Query $Q = \{q_1, q_2\}$
- Distance $D() = \text{Euclidean}$



Dominance check?
 $D(p_2, q_1) \leq D(p_1, q_1)$
AND
 $D(p_2, q_2) \leq D(p_1, q_2)$

For each point p_i
iterate over points p_j
if no point spatially
dominates p_i then add p_i to
spatial skyline

Time Complexity: $O(|P|^2 |Q|)$

$|P|$: number of data points, $|Q|$: number of query points

Problem Definition

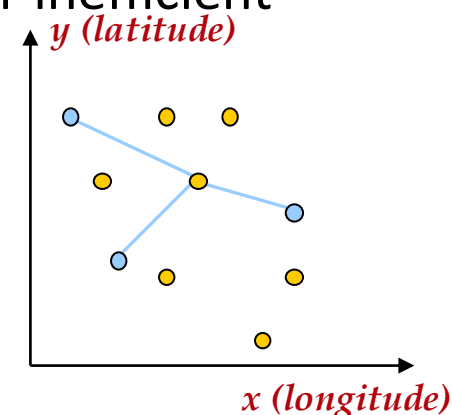
- **Naïve approach**

- **Complexity:** $O(|P|^2 |Q|)$

$|P|$: number of data points, $|Q|$: number of query points

- **Why a new algorithm is needed:**

- Complexity of Naïve approach is high
 - Each dominance check involves $2|Q|$ distance computation operations: increases with more query points
- General skyline algorithms are either inapplicable or inefficient
 - Due to dynamic spatial attributes
- Optimization opportunity
 - The geometric properties of space can be exploited





Geometric Properties

- **Complexity of Naïve approach:** $O(|P|^2 |Q|)$
 - $|P|$: number of data points
 - $|Q|$: number of query points
- We identify geometric properties to reduce this complexity by reducing the number of :
 - data points to be investigated
 - query points that has no effect on the result
- Less and cheaper dominance checks
- We identify three properties ...

Preliminaries: Voronoi Diagrams

- Given a set of spatial objects, a Voronoi diagram *uniquely* partitions the space into disjoint regions (cells).
- The region including object p includes all locations which are closer to p than to any other object p' .

Ordinary Voronoi Diagram

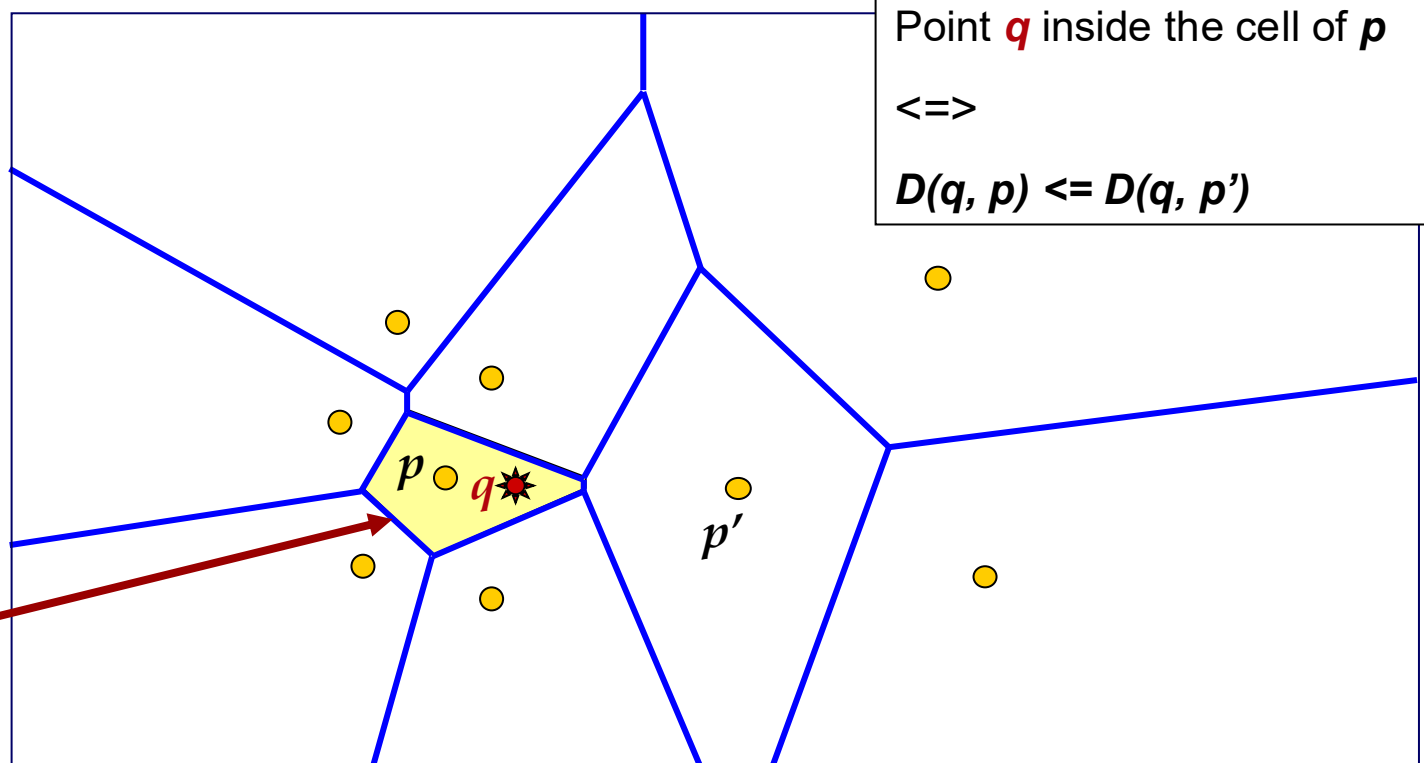
Dataset:

Points

Distance $D(.,.)$:

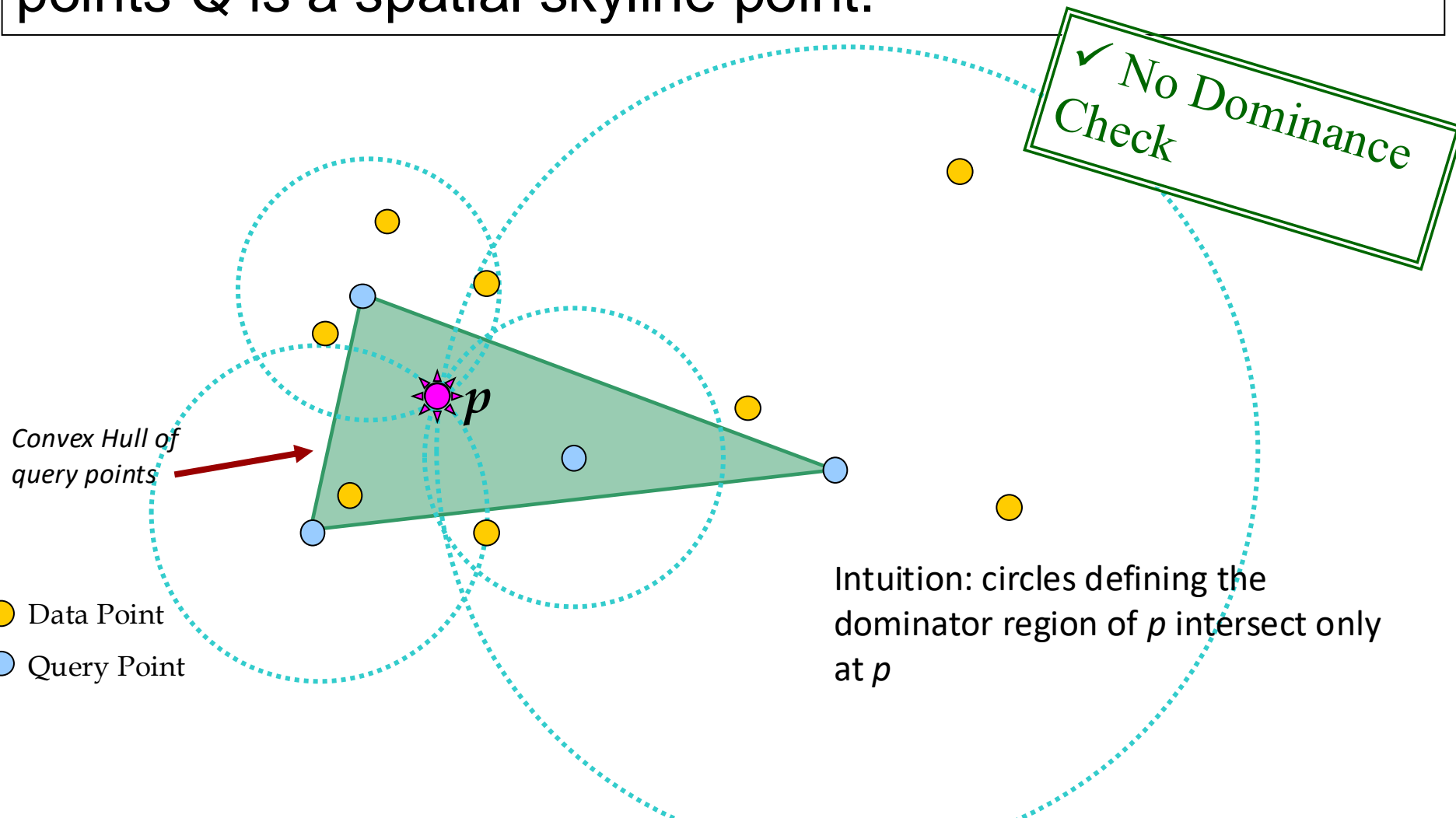
Euclidean (L_2)

Voronoi
Cell of p



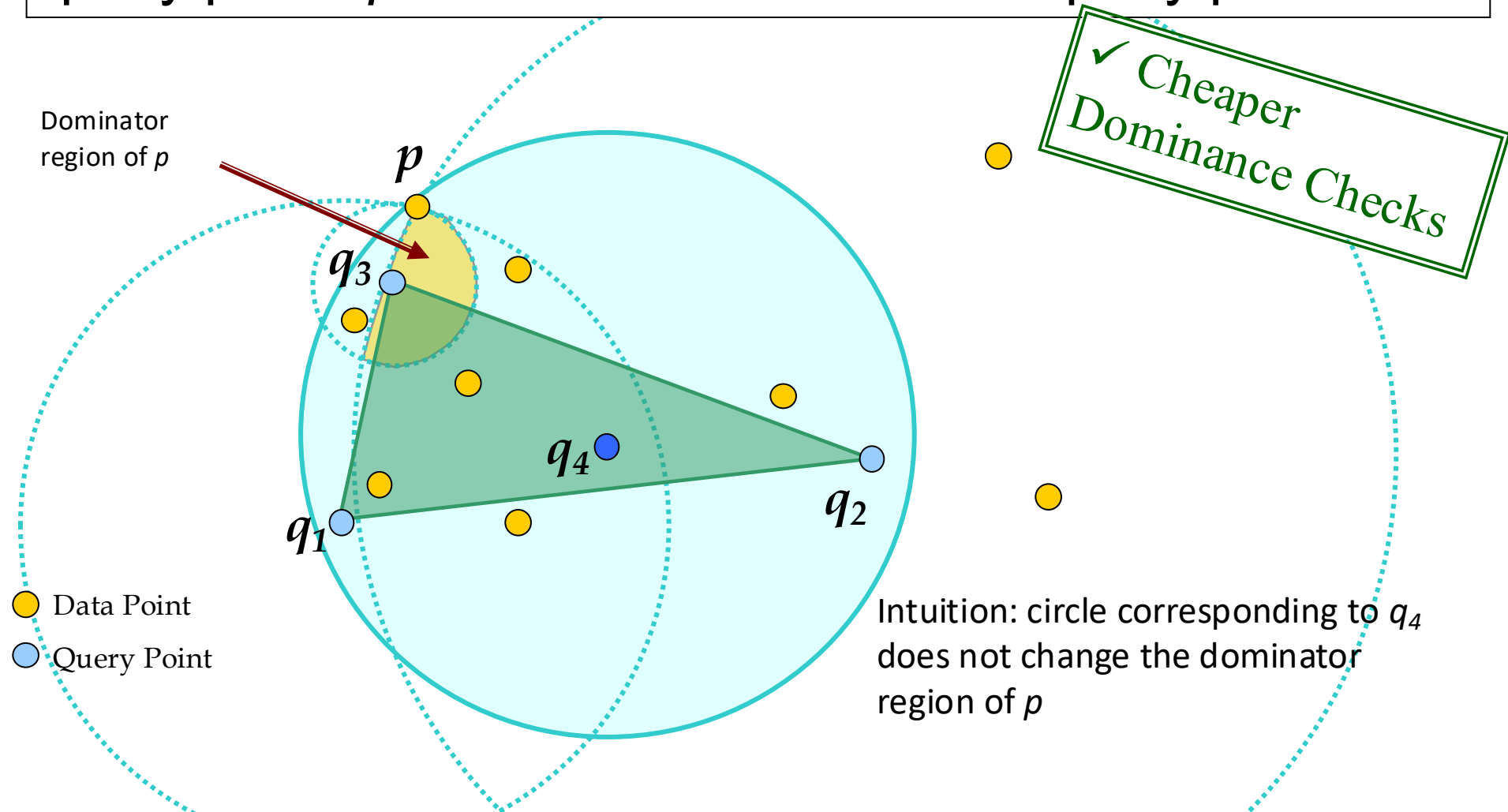
Geometric Properties

GP₁: Any point p inside the **convex hull** of query points Q is a spatial skyline point.



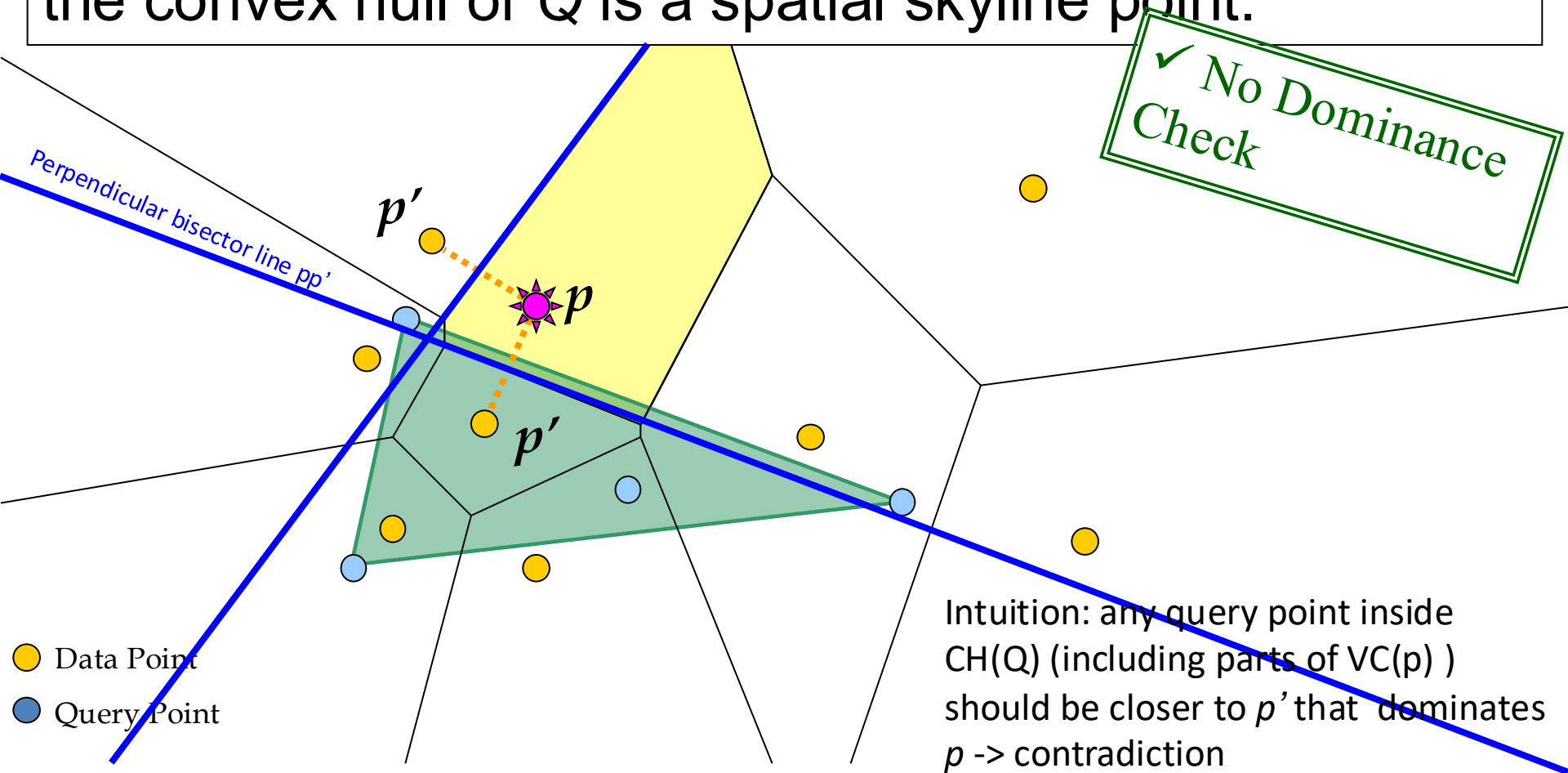
Geometric Properties

GP₂: The set of skyline points does not depend on any query point q inside the convex hull of query points Q .



Geometric Properties

GP₃: Any point p whose Voronoi cell **intersects** with the convex hull of Q is a spatial skyline point.



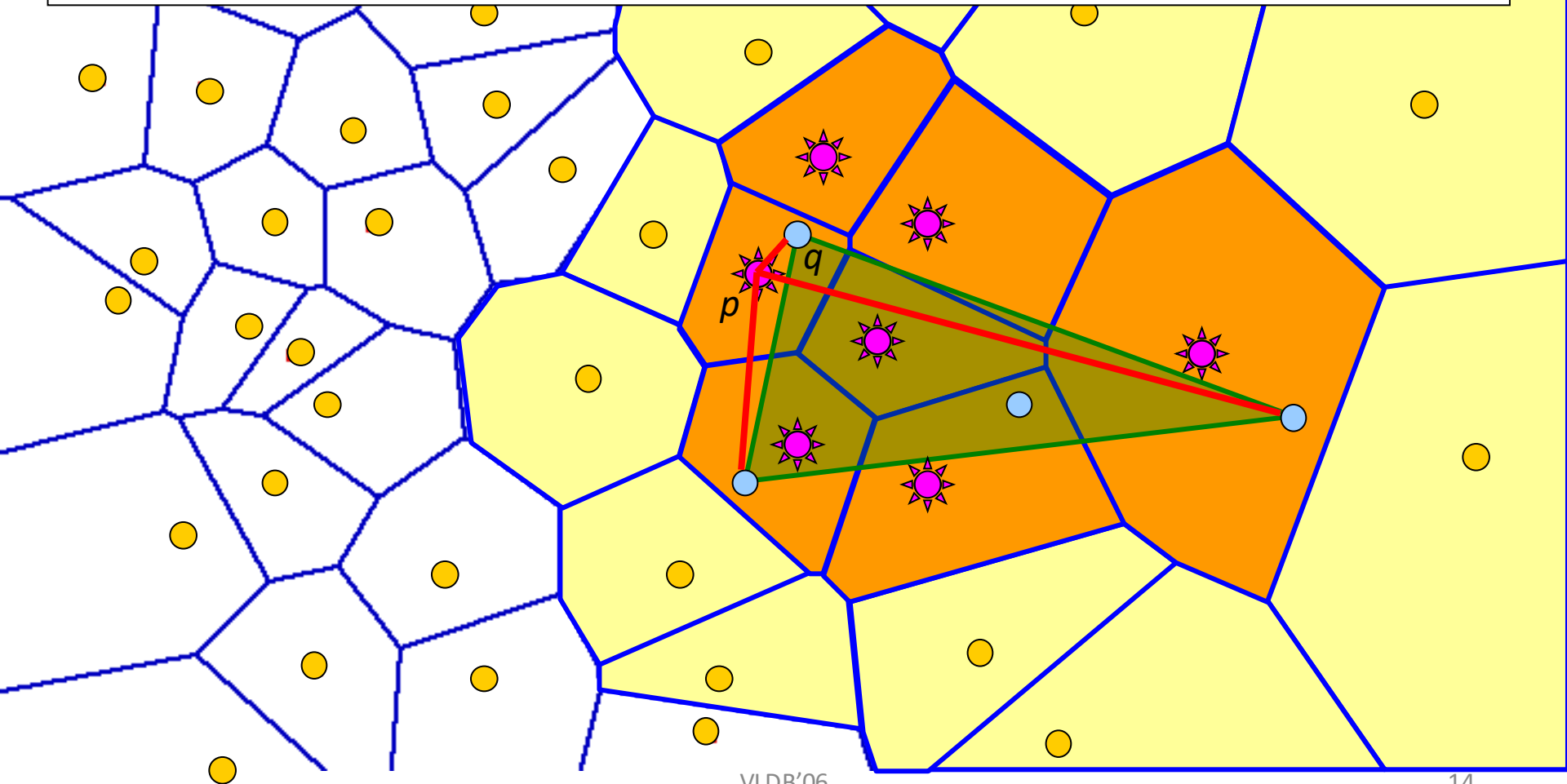
Algorithm: VS²

- **VS²**: Voronoi-based Spatial Skyline Algorithm
- Utilizes the geometric interpretation of the skyline
 - With no dominance check, adds any data point p whose Voronoi cell intersects with the convex hull of Q
GP₁
GP₃
 - Performs cheaper dominance check only on a small subset of points (neighbors of skyline points $\sim O(S)$)
GP₂
- Traverses the **Voronoi Diagram*** of data points

* Delaunay Graph

Algorithm: VS²

- We check the top of heap when all of its neighbors are already in the heap.
- No dominance check so far ...
- Check with only the current spatial skyline points



Algorithm: VS²

- **Time Complexity:** $O(|S|^2 |CH_v(Q)| + \Phi(|P|))$
 - Naïve: $O(|P|^2 |Q|)$
- $|S|$: number of skyline points
- $|CH_v(Q)|$: number of vertices of the convex hull of Q ($\leq |Q|$)
- $\Phi(|P|)$: complexity of finding the data point from which VS² starts traversing inside the convex hull of Q ($O(\log(|P|))$ with point location or $O(|P|^{1/2})$)
- **Space Complexity:** $O(|P|)$
 - Space required for ordinary Voronoi Diagram is $O(|P|)$

Algorithms: B²S²

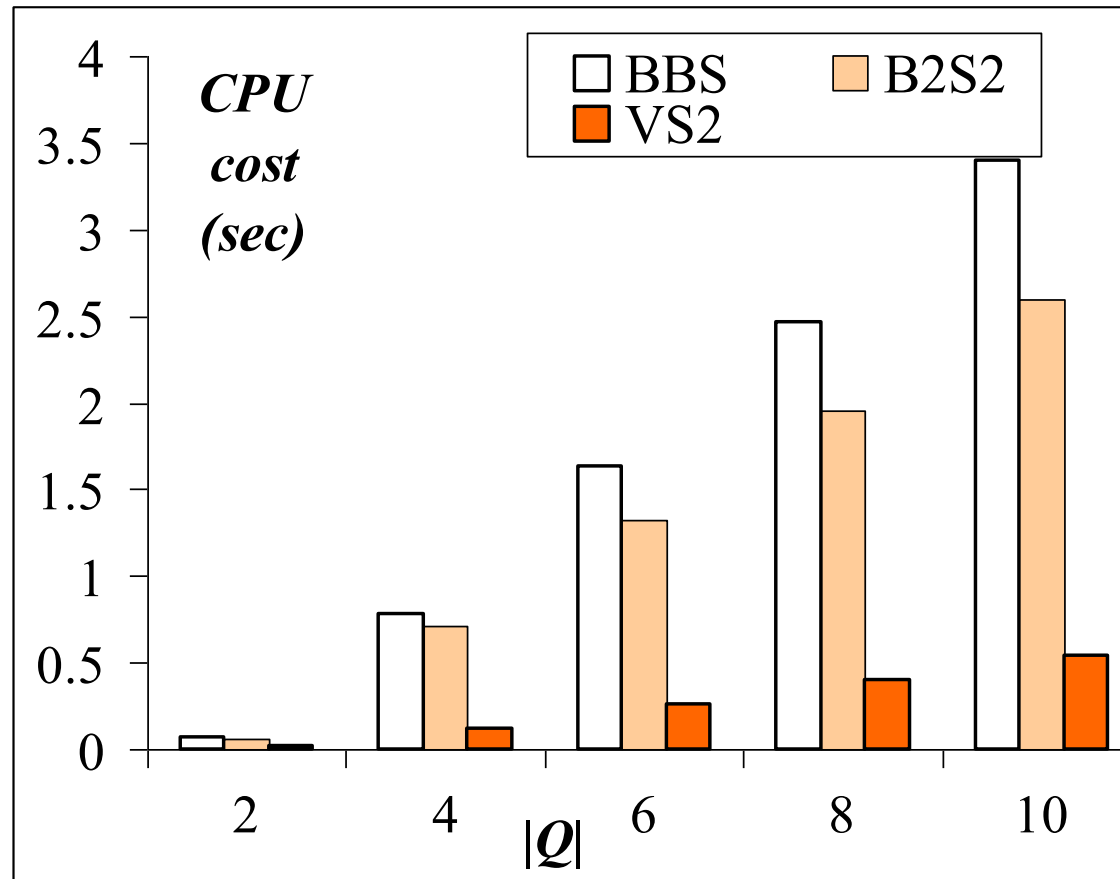
- **B²S²**: Branch-and-Bound Spatial Skyline Algorithm
- Customization of BBS [Papadias et al.] for SSQs
- Uses some of the geometric properties of the skyline (GP₁ and GP₂)
- Similar to BBS traverses an R-tree on data points
- **Traversal order**: specified by any monotone function (e.g., $\text{mindist}(p, CH_v(Q))$)



Performance Evaluation

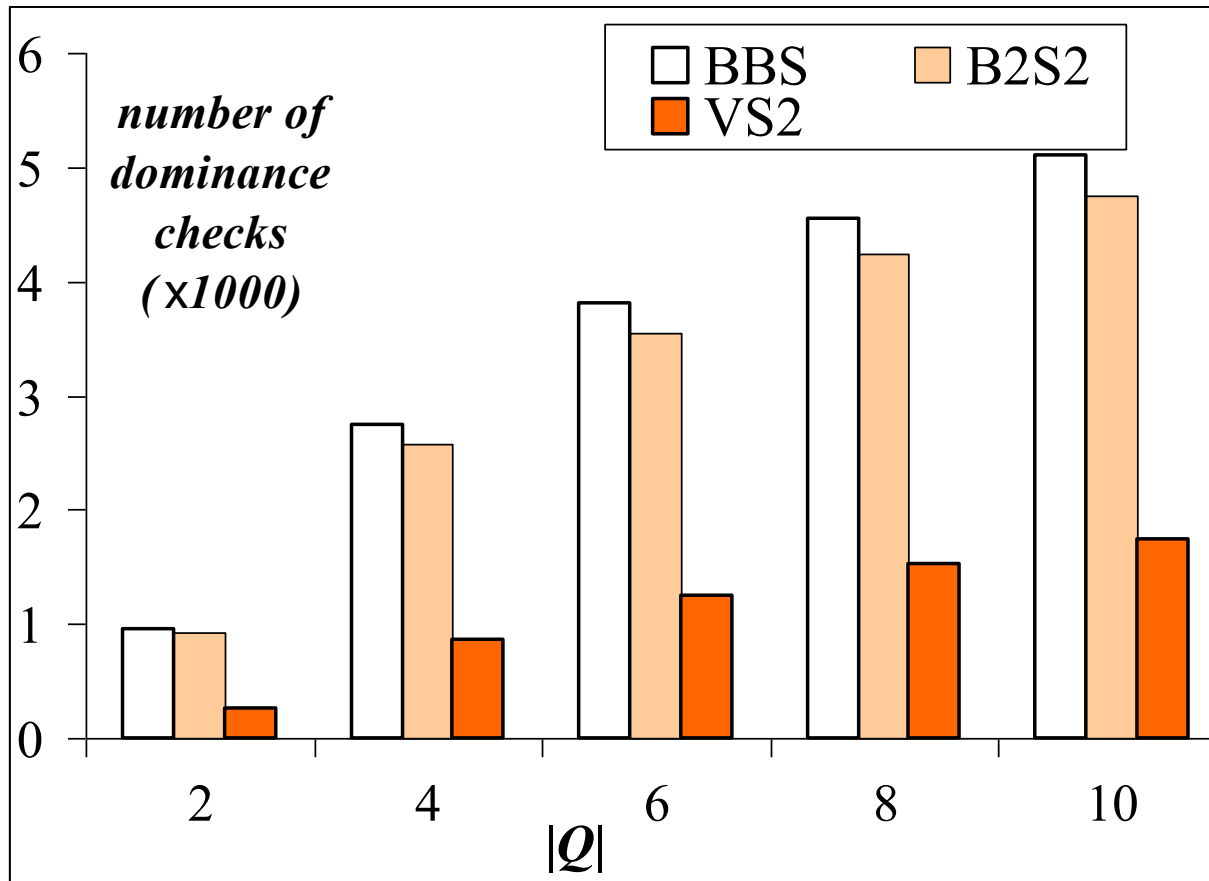
- **Dataset:** USGS including one million locations
- R^* -tree on data points for BBS and B^2S^2
- Pre-built Delaunay graph of data points for VS^2

Performance Evaluation



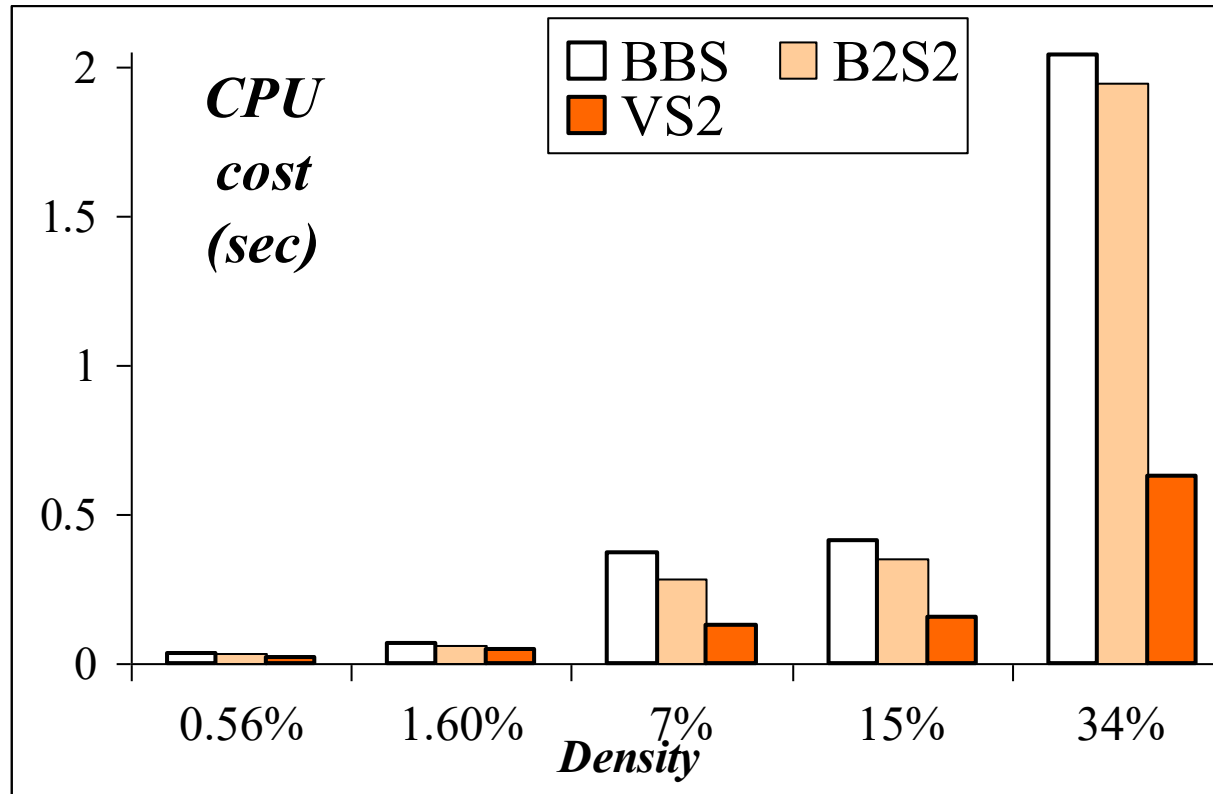
- $Max MBR(Q)=0.3\%$
- The difference in improvement of VS^2 over BBS increases for larger query sets.

Performance Evaluation



- Variations of B²S² require less dominance checks than BBS.
- Note that each dominance check is cheaper in our VS² and B²S² algorithms.

Performance Evaluation



- $\text{Max } |MBR(Q)| = 0.5\%$, $|Q| = 6$
- VS² is also scalable with respect to the density of data (i.e., number of skyline points)



Conclusion and Future Work

- We introduced the spatial skyline queries.
- We exploited the geometric properties of its solution space.
- We proposed two algorithms:
 - B^2S^2 that uses our properties to customize BBS for SSQs
 - VS^2 that utilizes a Voronoi diagram to minimize the number of dominance checks
- We proposed two variations of VS^2 for:
 - continuous spatial skyline query
 - handling non-spatial attributes
- VS^2 significantly outperforms its competitor approach BBS.

Future Work

- Addressing SSQ in other spaces
- Studying variations of SSQ



References

- Mehdi Sharifzadeh and Cyrus Shahabi, The Spatial Skyline Queries, VLDB 2006, Seoul, Korea, September 2006.